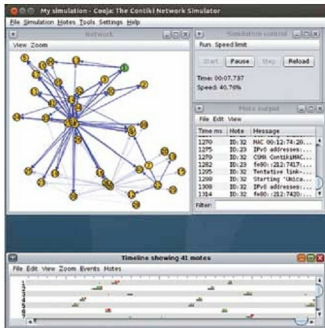


Contiki: The Operating System for Internet of Things

Contiki is an operating system just like Microsoft Windows and Linux but for a very specific purpose and mainly focused on 'things' in the Internet of Things (IoT). Here is everything you need to know about it



Screenshot of Contiki 2.6 running on 41 nodes forming an IPv6/RPL/6lowpan network in the Cooja Contiki network simulator (Courtesy: en.wikipedia.org)

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The Internet of Things (IoT), a technology that enables the automated exchange of key information between machines, and then ultimately to humans, promises a future where a whopping 50 billion devices will have the ability to "talk to each other" by 2020. The 'things' that are getting connected vary significantly in size and shape include tiny wireless sensors (smart dust) to electrical appliances at home to electronics used in space experiments. Things are characterised as low-memory and low-power devices, where memory is measured in bits or bytes (not megabytes or gigabytes) and power consumption is so

low that devices can run for years with a pair of AA-size battery!

Things are also referred to as smart objects which collect valuable data from the surroundings and communicate to the networked system. Thereby smart objects become a bridge between the real and virtual worlds. They contain sensors, microprocessor, communication and power source. The operating system (OS) in these objects controls all of its operations.

While we may see the number (50 billion) questionable, there are technologies being developed to make this a reality. The fundamental piece for that is the OS such as Contiki. Just like most consumer electronics devices come with chips embedded in them, for Internet of Things it is essential that there is some basic software such as Contiki running in them.

What is Contiki?

Contiki is an OS just like Microsoft Windows and Linux, but for a very specific purpose and mainly focused on 'things' in the IoT. Various functions of an OS include program/process management, resource manage-

ment, memory management and communication management. The goal of Contiki is to meet requirements of the smallest devices such as smart dust. Things should be able to communicate a few bits of data to one another.

Contiki is an open source software project with the vision to create a minimalist and working OS for wide deployment. This project has broken many myths about the smallest footprint in which an OS can be stored and made to function. One such software is the uIPv6—world's smallest IPv6 stack—which is integrated into Contiki. While there are many similar OS such as TinyOS, what makes Contiki different is the completeness and flexibility it offers to the programmers.

Features and functions of Contiki

Contiki is a completely open source software, distributed using the 3-clause BSD-style licence. The complete code is available on github for use or further development. Infrastructure is in place to run nightly regression tests to ensure the integrity of the software.

Contiki comes with a rich set of features that are device- and programmer-friendly. This project is under active development by various developers all around the world, and the software also has consumers in companies like Cisco and Google. Contiki OS can fit into 10 kB of RAM and 100 kB of ROM. It can run on devices such as 8051 SoC to ARM-powered devices. Ports are available on other platforms such as Arduino and Atmel. While standard compliance could be needed to inter-